NP-Completeness Reductions (ADA 2023, CSE 222)

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Outline

1. Some NP-Completeness Reductions

2. Another problem in Network Flow

- **Problem-1:** Given a nonnegative edge-weighted directed graph G and two vertices s and t. Design an algorithm that determines if G have a path from s to t with total weight at most k for some $k \in \mathbb{N}$.
- **Problem-2:** Given a nonnegative edge-weighted directed graph G and two vertices s and t. Design an algorithm that determines an explicit path of total shortest weight from s to t in G.
- **Question:** How do we define the verification for the same problem?

In addition to G, s, t, a sequence of vertices is also given. $v_1, v_2, \ldots, v_r$

Verification algorithm: check if $s = v_1$, $t = v_r$ and $(v_1, v_2), (v_2, v_3), \ldots, (v_{r-1}, v_r)$ are the edges of G.

- **CNF-Satisfiability:** A boolean formula

$\phi = (x_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee \bar{x}_4) \wedge (\bar{x}_2 \vee x_3 \vee \bar{x}_4) \wedge (x_1 \vee \bar{x}_3 \vee x_4)$.

- How do we define verification of CNF-SAT?
- What is an input instance and what is a certificate?

3CNF

$x_i \in \{\text{true}, \text{false}\}$

In addition to ϕ, an assignment is given, e.g.

$\boxed{x_1 = \text{false},\ x_2 = \text{false},\ x_3 = \text{false},\ x_4 = \text{true}}$

$x_1 = \text{true},\ x_2 = \text{true},\ x_3 = \text{false},\ x_4 = \text{true}$

Every clause has 3 literals.

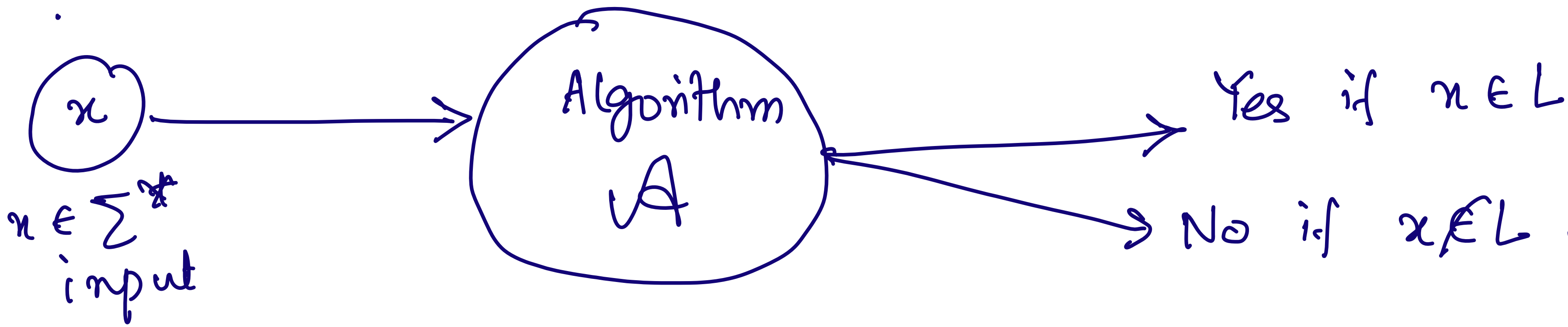
- **3-SAT:** A boolean formula φ of n variables $x_1, \ldots, x_n$ in *CONJUNCTIVE NORMAL FORM.* m clauses

3CNFSAT

- **Question:** Is there an assignment in $\{true, false\}^n$ of every variable x_i to *true, false* that evaluates φ to true?
- What is input instance and what is a certificate here?

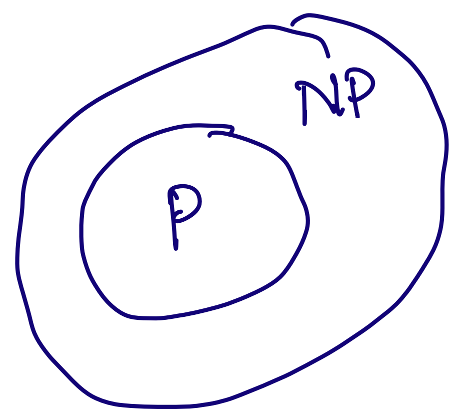
Certificate: An assignment of variables $\{true, false\}^n$

- **Class P:** Problems that can be solved in polynomial-time.

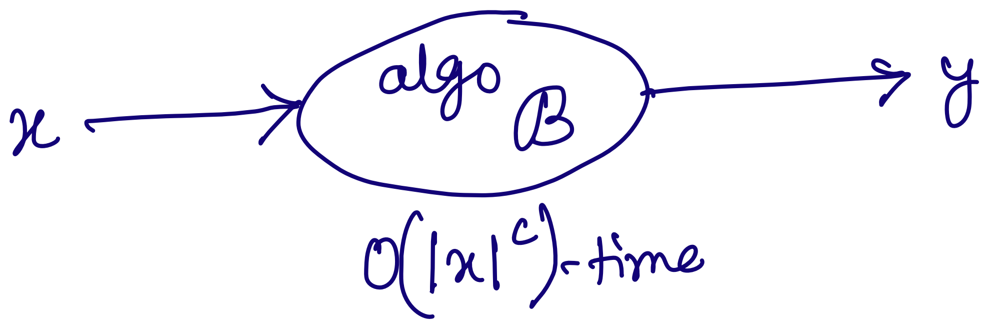


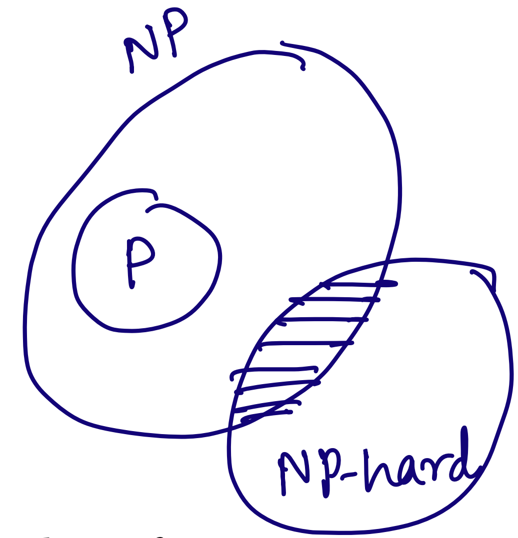
- **Class NP:** Problems the verification of which can be done in polynomial-time.

3SAT, INDEPENDENT SET



- **Polynomial time Reducibility:** A problem L_1 admits a *polynomial-time reduction* to a problem L_2 if given an input instance $x \in \Sigma^*$ of L_1, there is a polynomial-time algorithm that outputs an instance $y \in \Sigma^*$ of L_2 such that $x \in L_1$ if and only if $y \in L_2$. $L \leq_p L_2$



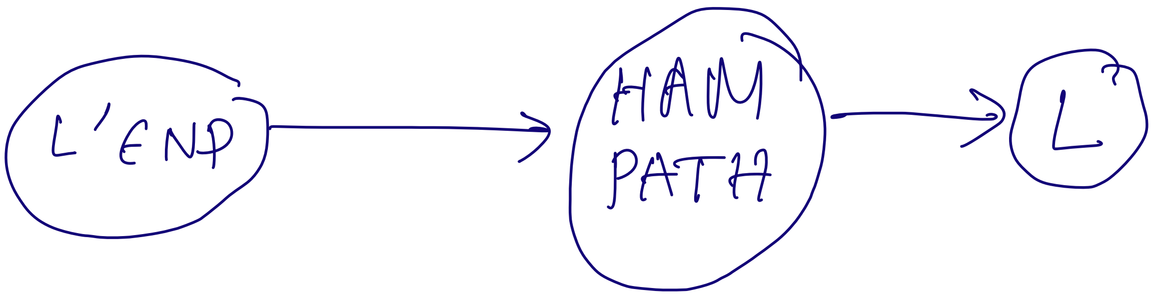


- **Class NP-Complete:** A problem L is said to be NP-Complete if
  - $L \in \mathsf{NP}$ and
  - For every $L' \in \mathsf{NP}$, there is a polynomial-time reduction from L' to L.

NP-hard

Suppose it is known that HAMILTONIAN PATH is NP-Complete.

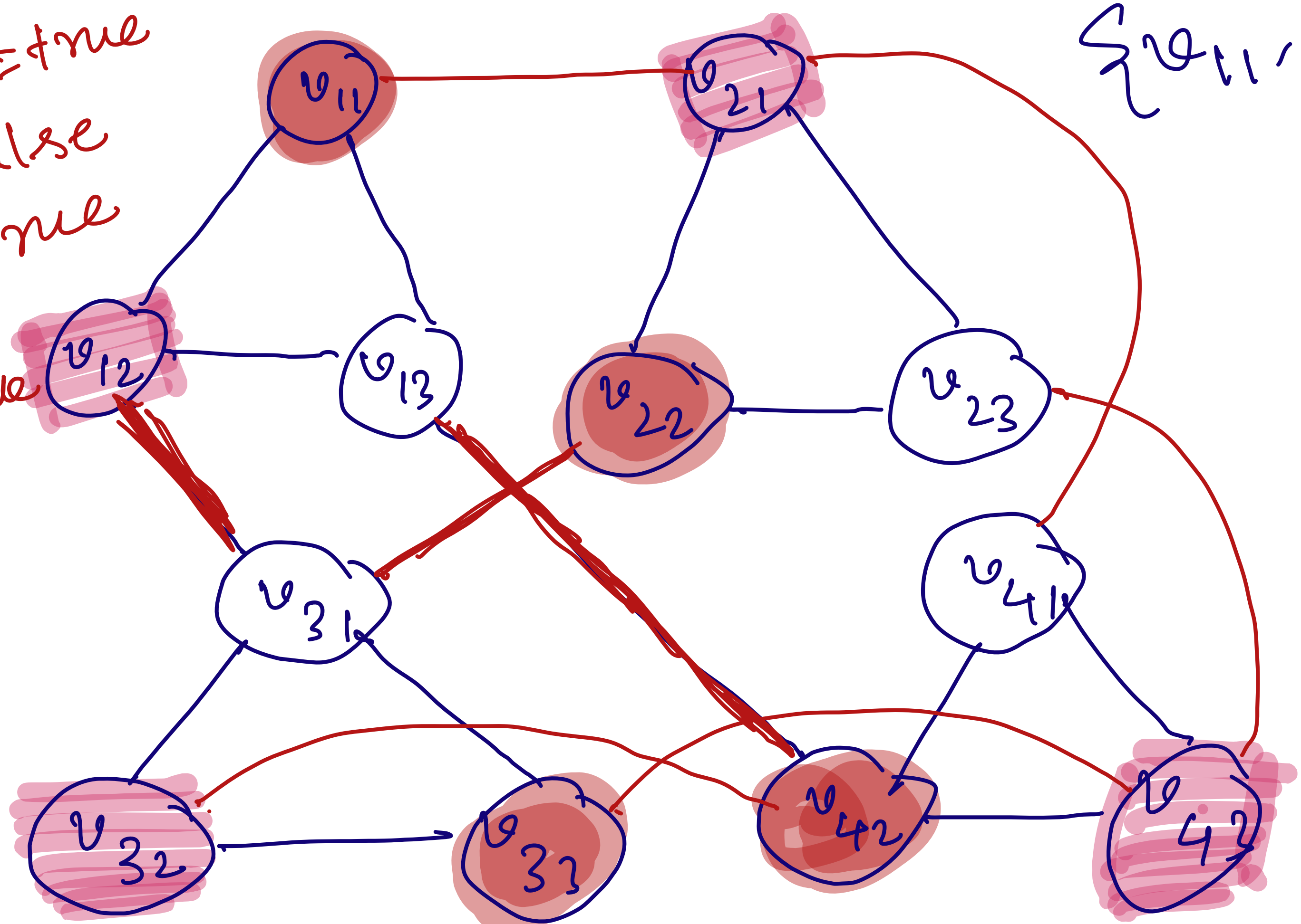
You design an algorithm B that gives a polynomial time reduction from HAMILTONIAN PATH to a problem L.



- Independent Set: Given a graph $G = (V, E)$ and $k \in \mathbb{N}$. Does G have a subset $S \subseteq V(G)$ with at least k vertices such that for every $u, v \in S$, (u, v) is not an edge in G.
- Reduction from 3-CNF-SAT to INDEPENDENT SET.
- **CNF-Satisfiability:** A boolean formula

$\phi =$ $(x_1 \vee x_2 \vee x_3) \wedge (\bar{x}_1 \vee x_2 \vee \bar{x}_4) \wedge (\bar{x}_2 \vee x_3 \vee \bar{x}_4) \wedge (x_1 \vee \bar{x}_3 \vee x_4)$.

$v_{12} = x_2 = \text{true}$
$x_1 = \text{false}$
$x_3 = \text{true}$
$x_4 = \text{true}$



$\{v_{11}, v_{22}, v_{42}, v_{33}\}$

G_ϕ

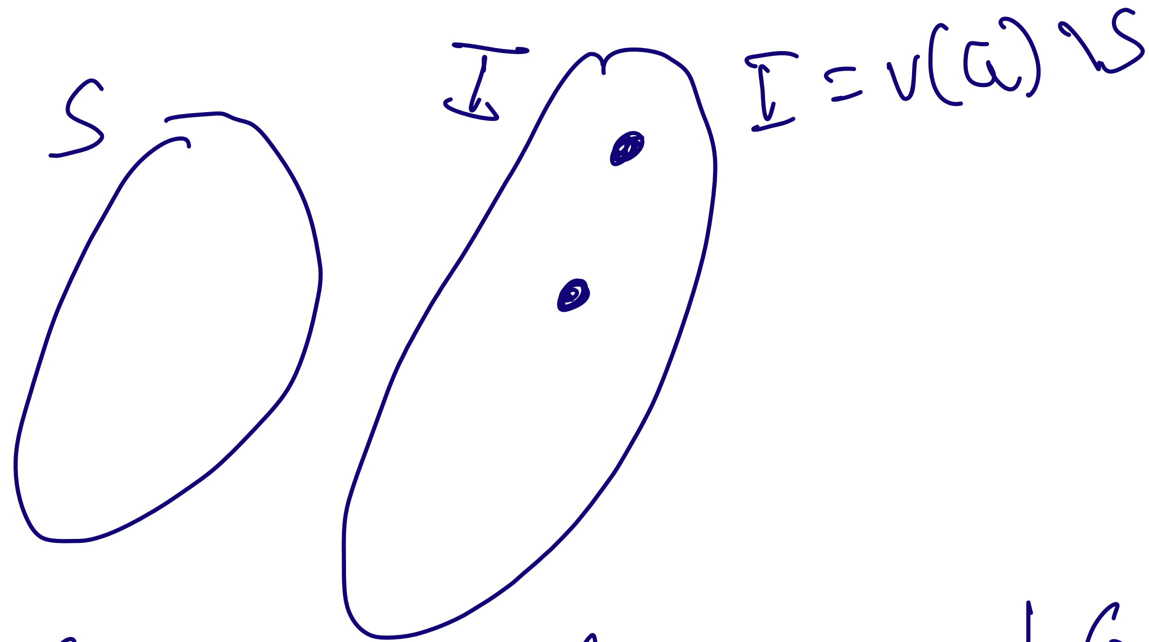
$x_1 = \text{true}$
$x_2 = \text{true}$
$x_3 = \text{false}$
$x_4 = \text{false}$

Lemma: ϕ is satisfiable if and only if G_ϕ has an independent set with at least m vertices ($m = 4$ = number of clauses)

($\Leftarrow$) Let S be an independent set of at least m vertices.

Then, from every triangle, at least one vertex is in S.

- **Vertex Cover:** Given a graph $G = (V, E)$ and $k \in \mathbb{N}$. Does G have a subset $S \subseteq V(G)$ with at most k vertices such that ~~for every $u, v \in S$, (u, v) is not an edge in G~~? for every $(u,v) \in E(G)$, $u \in S$ or $v \in S$?
- **Feedback Vertex Set:** Given a graph $G = (V, E)$ and $k \in \mathbb{N}$. Does G have a subset $S \subseteq V(G)$ with at most k vertices such that $G - S$ is acyclic?



A set $S \subseteq V(G)$ is a vertex cover of G, i.e. for every $(u,v) \in E(G)$, $u \in S$ or $v \in S$ if and only if $I = V(G) \setminus S$ is an independent set.

- **Vertex Cover:** Given a graph $G = (V, E)$ and $k \in \mathbb{N}$. Does G have a subset $S \subseteq V(G)$ with at most k vertices such that for every $u, v \in S$, (u, v) is not an edge in G?
- **Dominating Set:** Given a graph $G = (V, E)$ and $k \in \mathbb{N}$. Does G have a subset $S \subseteq V(G)$ with at most k vertices such that for every vertex v, either $v \in S$ or has a neighbor $u \in S$?

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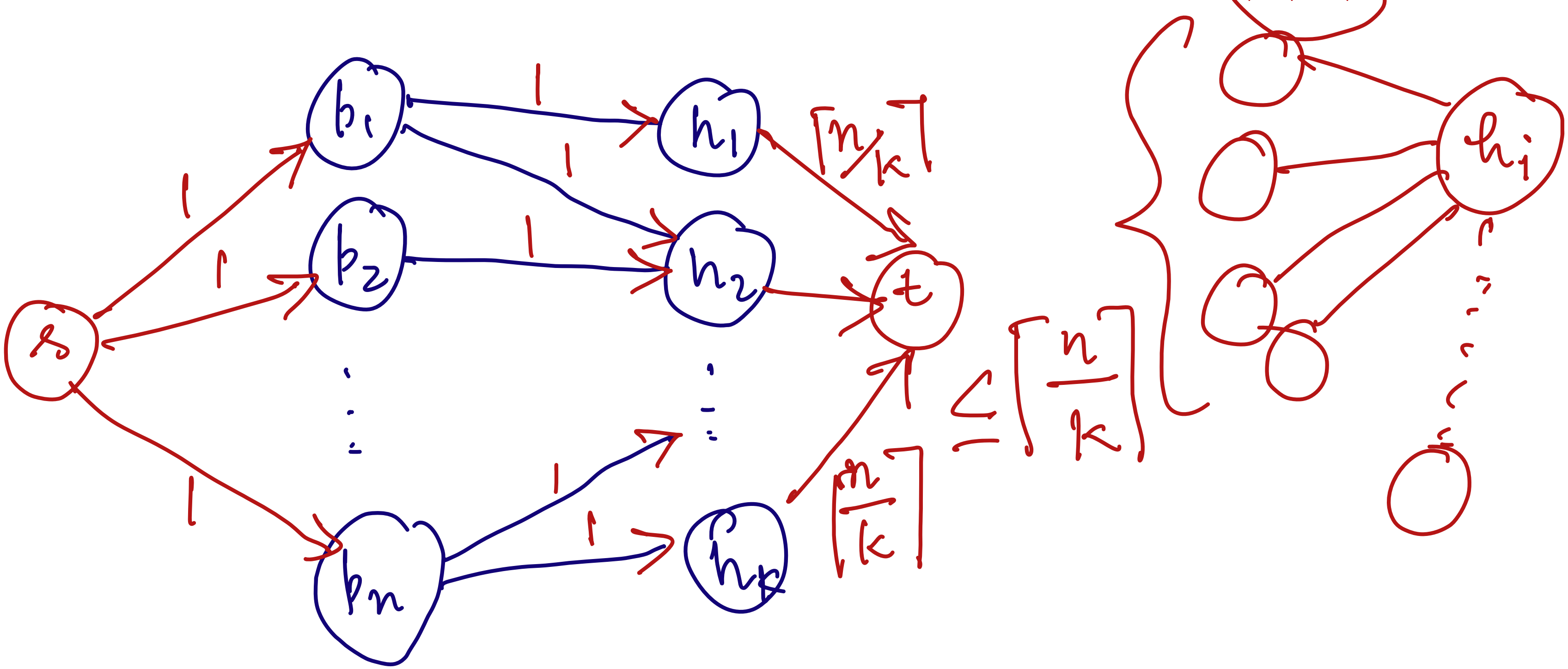
- Due to large-scale flooding in a region, paramedics have identified that there are n injured people and they have to be distributed across k hospitals in that region. An injured people have to be brought to a hospital that that is within 30 minutes driving distance from their current location. At the same time, one does not want to overload any hospital by sending too many patients its way. They want to work out whether each injured person can be taken to a hospital such that each hospital receives at most $\lceil n/k \rceil$ people.



Necessary and sufficient condition:

G has a flow from s to t of value n

if and only if

every injured person can be brought to some hospital and no hospital gets more than $\lceil n/k \rceil$ people.